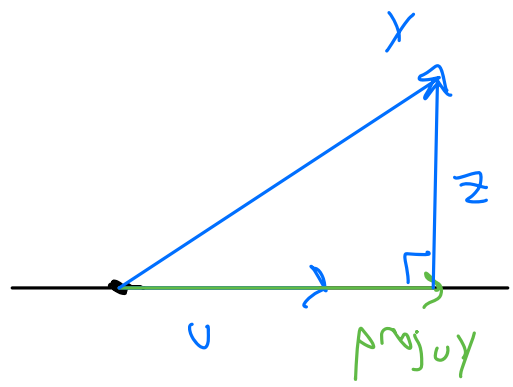
I am covering for Paul today.

<u>Recall</u> For u in $\mathbb{R}^n$ (or $\mathbb{C}^n$), $u \neq 0$,

any y in $\mathbb{R}^n$ (or $\mathbb{C}^n$), the

orthogonal projection of y onto u

is $$\text{proj}_u y = \frac{\langle y, u \rangle}{\langle u, u \rangle} u$$



and $y = \text{proj}_u y + z$

with $\langle z, u \rangle = 0$.

<u>Ex</u> Take $u = \begin{bmatrix} 2 \\ 1-i \end{bmatrix}$, $y = \begin{bmatrix} 1 \\ i \end{bmatrix}$ in $\mathbb{C}^2$.

Then $$\text{proj}_u y = \frac{\langle y, u \rangle}{\langle u, u \rangle} u$$

$$= \frac{(1)(\overline{2}) + (i)(\overline{1-i})}{2^2 + 1^2 + 1^2} \begin{bmatrix} 2 \\ 1-i \end{bmatrix}$$

$$= \frac{2 + i - 1}{6} \begin{bmatrix} 2 \\ 1-i \end{bmatrix}$$

$$= \frac{1+i}{6} \begin{bmatrix} 2 \\ 1-i \end{bmatrix}$$

Exercise: If we replace u by cu for any nonzero scalar c, $\text{proj}_{cu} y = \text{proj}_u y$.

Since $L = \{cu : c \in \mathbb{R} \text{ (or } \mathbb{C})\} = \text{span}\{u\}$ is a subspace, it makes sense to define the orthogonal projection of y onto L.

$$\text{proj}_L y = \frac{\langle y, u \rangle}{\langle u, u \rangle} u$$

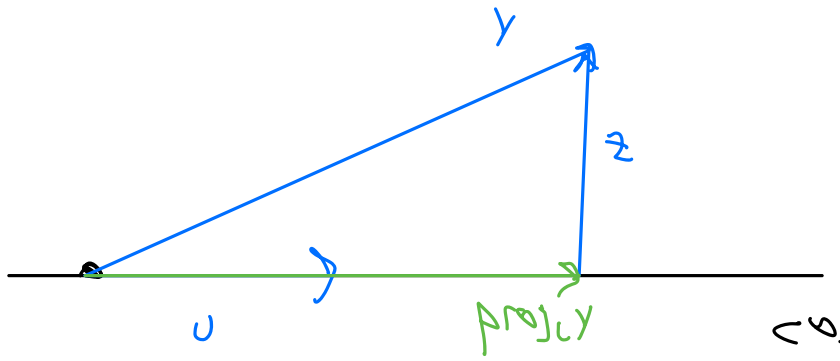
$$y = \text{proj}_L y + z \quad \text{where } z \in L^\perp.$$

<u>Ex</u> Let $L = \text{span}\left\{\begin{bmatrix}1\\1\\1\end{bmatrix}\right\} \subseteq \mathbb{R}^3$ and $y = \begin{bmatrix}2\\0\\-1\end{bmatrix}$.

Then $$\text{proj}_L y = \frac{\left\langle \begin{bmatrix}2\\0\\-1\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix} \right\rangle}{\left\langle \begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix} \right\rangle} \begin{bmatrix}1\\1\\1\end{bmatrix}$$

$$= \frac{2+0-1}{1+1+1} \begin{bmatrix}1\\1\\1\end{bmatrix} = \frac{1}{3}\begin{bmatrix}1\\1\\1\end{bmatrix}$$

<u>Some geometry</u>



$\text{proj}_L y$ is the vector that minimizes the vector z that connects y to L.

For this reason, we define the <u>distance of y to L</u> by

$$\|\underbrace{y - \text{proj}_L y}_{z \text{ in picture}}\|$$

In the previous example,

$$\left\| \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} - \frac{1}{3}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\| = \left\| \begin{bmatrix} 5/3 \\ -1/3 \\ -4/3 \end{bmatrix} \right\|$$

$$= \sqrt{\left(\frac{5}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(-\frac{4}{3}\right)^2}$$

$$= \sqrt{\frac{42}{9}}$$

<u>Def</u> A set $\{v_1, \dots, v_k\}$ in $\mathbb{R}^n$ (or $\mathbb{C}^n$) is

<u>orthonormal</u> if it is orthogonal and every v_i is a unit vector.

That is, if $\langle v_\ell, v_j \rangle = 0$ if $\ell \neq j$

$$\langle v_j, v_j \rangle = 1$$

<u>Ex</u> If we normalize

$$u_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \quad u_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \quad u_3 = \begin{bmatrix} -1 \\ -4 \\ 7 \end{bmatrix}$$

(which are orthogonal) from before, we obtain

$$v_1 = \frac{1}{\|u_1\|} u_1 = \frac{1}{\sqrt{3^2+1^2+1^2}} \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{11}} \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

$$v_2 = \frac{1}{\|u_2\|} u_2 = \frac{1}{\sqrt{6}} \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$$

$$v_3 = \frac{1}{\|u_3\|} u_3 = \frac{1}{\sqrt{66}} \begin{bmatrix} -1 \\ -4 \\ 7 \end{bmatrix}.$$

The resulting set $\{v_1, v_2, v_3\}$ is still

orthogonal (exercise), so it is orthonormal.

Let's check

$$\langle v_1, v_2 \rangle = \frac{1}{\sqrt{66}}(-3+2+1) = 0.$$

<u>Ex</u> $v_1 = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 \\ i \end{bmatrix}$ and $v_2 = \frac{1}{\sqrt{3}}\begin{bmatrix} i \\ 1 \end{bmatrix}$ form an orthonormal set in $\mathbb{C}^2$.

(check!)

<u>Def</u> A basis for a subspace of $\mathbb{R}^n$ (or $\mathbb{C}^n$) is called an <u>orthonormal basis</u> if it is an orthonormal set.

In the previous two examples, we can use Theorem 4 to show they are linearly independent. Use Thm 13 §4.5 to conclude that they are bases.

Theorem 5 revisited

Let $\{v_1, \dots, v_k\}$ be an <u>orthonormal</u> basis for a subspace W in $\mathbb{R}^n$ (or $\mathbb{C}^n$). Then $y \in W$ has the form

$$y = \langle y, v_1 \rangle v_1 + \langle y, v_2 \rangle v_2 + \dots + \langle y, v_k \rangle v_k .$$

<u>Sketch of proof</u> In original thm 5, coefficients were $\dfrac{\langle y, v_i \rangle}{\langle v_i, v_i \rangle}$. But $\langle v_i, v_i \rangle = 1$ since v_i is a unit vector.

<u>Ex</u> Take $y = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and $\{v_1, v_2, v_3\} \subseteq \mathbb{R}^3$ as before. Then

$$y = \left\langle \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{11}} \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} \right\rangle \left(\frac{1}{\sqrt{11}} \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} \right) + \left\langle \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{6}} \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \right\rangle \left(\frac{1}{\sqrt{6}} \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \right)$$

$$+ \left\langle \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{66}} \begin{bmatrix} -1 \\ -4 \\ 7 \end{bmatrix} \right\rangle \left(\frac{1}{\sqrt{66}} \begin{bmatrix} -1 \\ -4 \\ 7 \end{bmatrix} \right)$$

$$= \frac{3}{11}\begin{bmatrix}3\\1\\1\end{bmatrix} + \left(\frac{-1}{6}\right)\begin{bmatrix}-1\\2\\1\end{bmatrix} + \left(\frac{-1}{66}\right)\begin{bmatrix}-1\\-4\\7\end{bmatrix}$$

<u>Ex</u> Take $y = \begin{bmatrix}1\\-1\end{bmatrix} \in \mathbb{C}^2$, $v_1 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\i\end{bmatrix}$, $v_2 = \frac{1}{\sqrt{2}}\begin{bmatrix}i\\1\end{bmatrix}$

$$y = \left\langle \begin{bmatrix}1\\-1\end{bmatrix}, \frac{1}{\sqrt{2}}\begin{bmatrix}1\\i\end{bmatrix}\right\rangle \left(\frac{1}{\sqrt{2}}\begin{bmatrix}1\\i\end{bmatrix}\right) + \left\langle \begin{bmatrix}1\\-1\end{bmatrix}, \frac{1}{\sqrt{2}}\begin{bmatrix}i\\1\end{bmatrix}\right\rangle \left(\frac{1}{\sqrt{2}}\begin{bmatrix}i\\1\end{bmatrix}\right)$$

$$= \frac{1}{2}(1+i)\begin{bmatrix}1\\i\end{bmatrix} + \frac{1}{2}(-1-i)\begin{bmatrix}i\\1\end{bmatrix}$$

<u>§6.3 Orthogonal Projections</u>

Recall that we argued if u is nonzero and $L = \text{span}\{u\}$ then

$$y = \text{proj}_L y + z = \frac{\langle y, u\rangle}{\langle u, u\rangle} u + z$$

where $z \in L^\perp$. Here L is a 1-dimensional subspace of $\mathbb{R}^n$ (or $\mathbb{C}^n$) with basis $\{u\}$. We can imitate this to define the projection

onto any finite dimensional subspace W.

<u>Def</u> Let W be a subspace of $\mathbb{R}^n$ (or $\mathbb{C}^n$) with orthogonal basis $\{w_1, \dots, w_k\}$.

The orthogonal projection of y onto W is

$$\text{proj}_W y = \frac{\langle y, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 + \dots + \frac{\langle y, w_k \rangle}{\langle w_k, w_k \rangle} w_k$$

<u>Ex</u> Let $w_1 = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$, $w_2 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$ and $y = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ in $\mathbb{R}^3$.

Let $W = \text{span}\{w_1, w_2\}$.

Note that $\langle w_1, w_2 \rangle = 0$ and $\{w_1, w_2\}$ is lin. indep (check).

$\{w_1, w_2\}$ is an orthogonal basis for W.

We compute:

$$\text{proj}_W y = \frac{\left\langle \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} \right\rangle}{\left\langle \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} \right\rangle} \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} + \frac{\left\langle \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \right\rangle}{\left\langle \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \right\rangle} \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

$$= \frac{9}{30} \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} + \frac{3}{6} \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix}$$

One can show that $v_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$ is also an orthogonal basis for W.

With this basis,

$$\text{proj}_W y = \frac{\left\langle \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\rangle}{\left\langle \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\rangle} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \frac{\left\langle \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} \right\rangle}{\left\langle \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} \right\rangle} \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$$

$$= 2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - \frac{1}{5} \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix}, \text{ as before.}$$